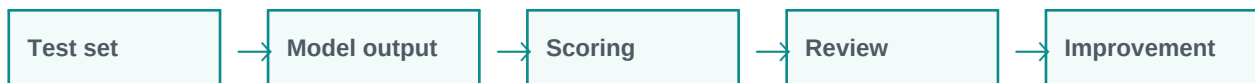


LLM Evaluation

Evaluation checks quality before and after deployment.

Simple explanation: LLM evaluation means checking whether an AI system is accurate, useful, safe, grounded, consistent, and suitable for production use.

Learning style: this handout starts with a very simple explanation and gradually increases the depth. It is designed for classroom training, participant reading, and quick revision.



What you will learn

- Meaning in simple words
- Key components and architecture
- Practical examples and use cases
- Where it fits in GenAI and agentic AI systems
- Benefits, challenges, and implementation points



1. Beginner-Friendly Explanation

LLM evaluation means checking whether an AI system is accurate, useful, safe, grounded, consistent, and suitable for production use.

Think of it like this:

Evaluation is like quality checking. It tells whether the AI answer is correct, safe, useful, and grounded.

Simple real-time examples

- Compare prompts
- Validate RAG answers
- Measure hallucination
- Review support replies
- Check classification quality

2. Gradually Deeper Explanation

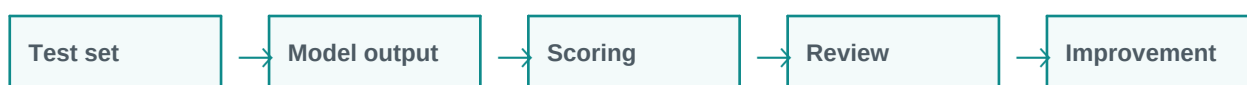
LLM Evaluation becomes more powerful when we understand its components, data flow, limitations, and business fit. In real projects, the concept is rarely used alone. It usually works with prompts, models, tools, documents, APIs, evaluation, monitoring, and human review.

Important concepts

Concept	Meaning
Accuracy	Accuracy is an important part of understanding and applying LLM Evaluation in practical GenAI systems.
Groundedness	Groundedness is an important part of understanding and applying LLM Evaluation in practical GenAI systems.
Relevance	Relevance is an important part of understanding and applying LLM Evaluation in practical GenAI systems.
Safety	Safety is an important part of understanding and applying LLM Evaluation in practical GenAI systems.
Latency/cost	Latency/cost is an important part of understanding and applying LLM Evaluation in practical GenAI systems.

3. Architecture / Flow

A typical architecture can be understood as a simple flow. The exact design changes based on the project, but the pattern below is a useful starting point.



Architecture explanation

Step 1: Test set - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Step 2: Model output - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Step 3: Scoring - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Step 4: Review - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Step 5: Improvement - This stage contributes to the final outcome. It should be designed carefully because weak inputs, wrong retrieval, poor prompts, or missing validation can reduce quality.

Common implementation layers

- User interface or API layer
- Prompt and instruction layer
- Model layer
- Tool/retrieval/data layer
- Validation, safety, and logging layer
- Human review or feedback layer

4. Detailed Examples

Below are example scenarios showing how this topic appears in real GenAI projects.

Scenario	How it works	Expected output
Compare prompts	The system receives the request, applies LLM Evaluation, and processes the task step by step.	A useful, structured, reviewable result.
Validate RAG answers	The system receives the request, applies LLM Evaluation, and processes the task step by step.	A useful, structured, reviewable result.
Measure hallucination	The system receives the request, applies LLM Evaluation, and processes the task step by step.	A useful, structured, reviewable result.
Review support replies	The system receives the request, applies LLM Evaluation, and processes the task step by step.	A useful, structured, reviewable result.



Scenario	How it works	Expected output
Check classification quality	The system receives the request, applies LLM Evaluation, and processes the task step by step.	A useful, structured, reviewable result.

5. Benefits and Challenges

Benefits	Challenges / Risks
Improves productivity and reduces repetitive manual effort.	Can produce incorrect or unsupported answers if not validated.
Helps users interact with complex systems in natural language.	Needs careful design of prompts, data access, permissions, and monitoring.
Can support training, support, coding, analysis, and document workflows.	May have cost, latency, privacy, and governance concerns in production.

6. Best Practices

- Start with a simple prototype before building a complex system.
- Use clear prompts and structured output formats.
- Add validation and human review for important decisions.
- Measure quality with realistic test cases, not only demo examples.
- Monitor cost, latency, accuracy, safety, and user feedback.
- Document assumptions, limitations, and escalation paths.

7. Key Takeaways

LLM Evaluation is an important concept in modern LLM and GenAI systems. The simple idea is easy to understand, but production implementation requires thoughtful architecture, data quality, evaluation, safety, and monitoring.

Quick revision

- Simple meaning: LLM evaluation means checking whether an AI system is accurate, useful, safe, grounded, consistent, and suitable for production use.
- Architecture: input, processing, tools/data, validation, output.
- Business value: saves time, improves knowledge access, and supports automation.
- Risk control: use grounding, evaluation, guardrails, monitoring, and human approval.